

Báo cáo thực hành backend 2

1. Code

- database.js

```
JS database.js X
src > config > JS database.js > ...
1 import mongoose from "mongoose";
2
3 export const connectDatabase = async (URI) => {
4   mongoose.connection.on("connected", () => console.log("MongoDB connected"));
5   mongoose.connection.on("error", (error) => console.log("MongoDB error:", error));
6   await mongoose.connect(URI);
7 };
```

- index.js

```
JS database.js JS index.js X
JS index.js > app.listen() callback
1 import "dotenv/config"
2
3 import app from "../src/app.js"
4 import { connectDatabase } from "../src/config/database.js";
5
6 const PORT = process.env.PORT || 3000;
7 const URI = process.env.URI;
8
9 await connectDatabase(URI);
10 app.listen(PORT, () => {
11   console.log(`Server is running on port ${PORT}...`);
12 })
```

- contact.service.js

```
JS database.js JS index.js JS contact.service.js X
src > services > JS contact.service.js > deleteOne
1 import Contact from "../models/contact.model.js"
2
3 export const create = (payload) => Contact.create(payload);
4
5 export const findAll = () => Contact.find();
6
7 export const deleteAll = () => Contact.deleteMany();
8
9 export const findAllFavorite = () => Contact.find({favorite: true});
10
11 export const findOne = (id) => Contact.findById(id);
12
13 export const update = (id, payload) => Contact.findByIdAndUpdate(id, payload, {new: true, runValidators: true}); // validate new payload and return new document
14
15 export const deleteOne = (id) => Contact.findByIdAndDelete(id);
```

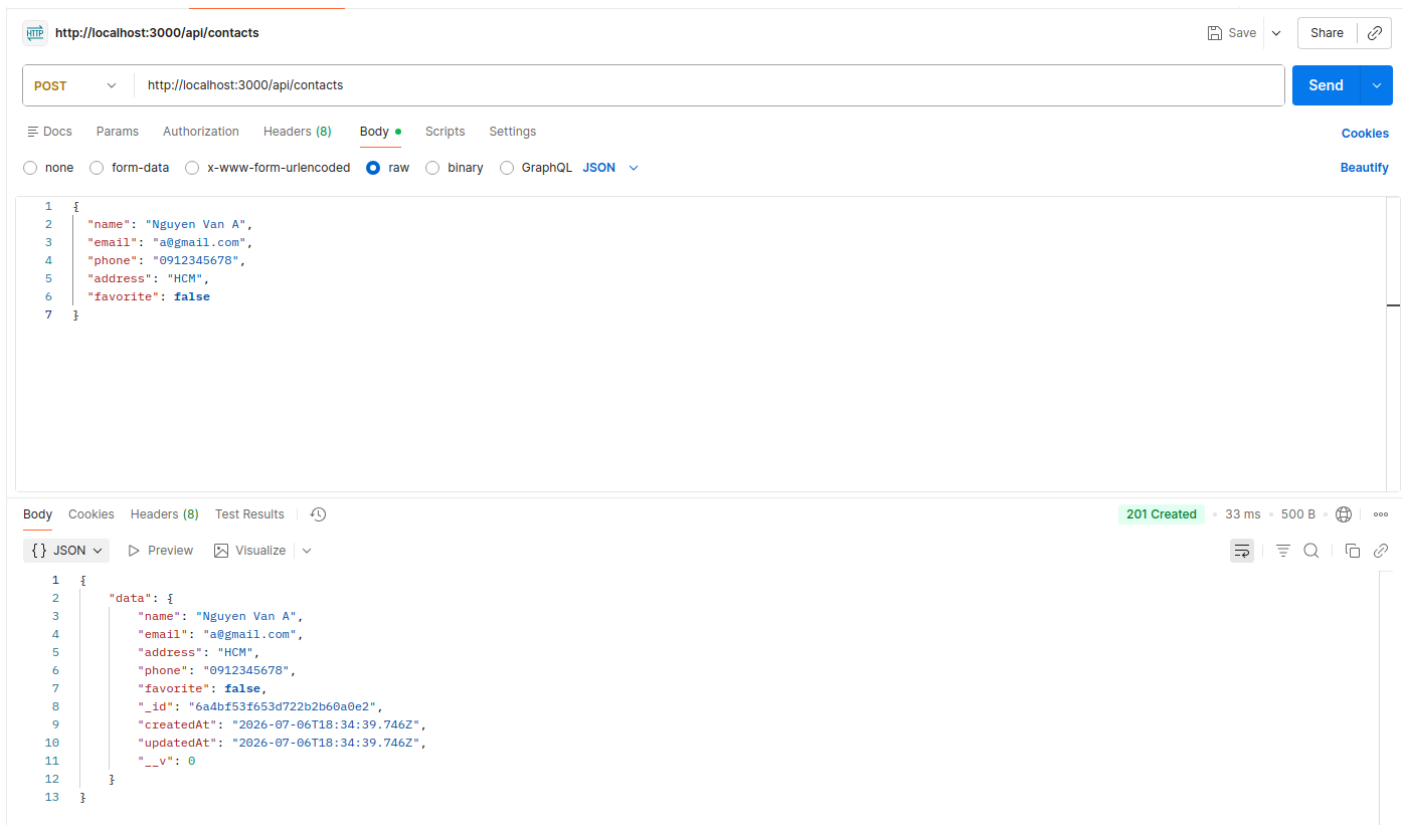
- contact.controller.js

```
JS database.js JS contact.controller.js X
src > controllers > JS contact.controller.js > findOne
1 import * as contactService from "../services/contact.service.js"
2
3 import {ApiError} from "../middlewares/error.middleware.js"
4
5 export const create = async (request, response) => {
6   const payload = request.body
7   const responseContent = await contactService.create(payload)
8   response.status(201).json({data: responseContent});
9 }
10
11 export const findAll = async (request, response) => {
12   const contacts = await contactService.findAll();
13   response.status(200).json({data: contacts});
14 }
15
16 export const deleteAll = async (request, response) => {
17   await contactService.deleteAll();
18   response.status(204).send();
19 }
20
21 export const findAllFavorite = async (request, response) => {
22   const favoriteContacts = await contactService.findAllFavorite();
23   response.status(200).json({data: favoriteContacts});
24 }
25
26 export const findOne = async (request, response) => {
27   const id = request.params.id;
28   const contact = await contactService.findOne(id)
29   if (!contact) {
30     throw new ApiError(404, "Contact not found");
31   }
32   response.status(200).json({data: contact});
33 }
34
35 export const update = async (request, response) => {
36   const id = request.params.id;
37   const payload = request.body;
38   const contact = await contactService.update(id, payload);
39   if (!contact) {
40     throw new ApiError(404, "Contact not found");
41   }
42   response.status(200).json({data: contact});
43 }
44
45 export const deleteOne = async (request, response) => {
```

```
JS database.js JS contact.controller.js X
src > controllers > JS contact.controller.js > findOne
17
20
21 export const findAllFavorite = async (request, response) => {
22   const favoriteContacts = await contactService.findAllFavorite();
23   response.status(200).json({data: favoriteContacts});
24 }
25
26 export const findOne = async (request, response) => {
27   const id = request.params.id;
28   const contact = await contactService.findOne(id)
29   if (!contact) {
30     throw new ApiError(404, "Contact not found");
31   }
32   response.status(200).json({data: contact});
33 }
34
35 export const update = async (request, response) => {
36   const id = request.params.id;
37   const payload = request.body;
38   const contact = await contactService.update(id, payload);
39   if (!contact) {
40     throw new ApiError(404, "Contact not found");
41   }
42   response.status(200).json({data: contact});
43 }
44
45 export const deleteOne = async (request, response) => {
46   const id = request.params.id;
47   await contactService.deleteOne(id);
48   response.status(204).send();
49 }
50
51
```

2. Test trên Postman

- POST <http://localhost:3000/api/contacts>



- GET <http://localhost:3000/api/contacts>

http://localhost:3000/api/contacts

GET http://localhost:3000/api/contacts

Docs Params Authorization Headers (6) Body Scripts Settings

none form-data x-www-form-urlencoded raw binary GraphQL JSON

1 Ctrl+Alt+P to Ask AI

Body Cookies Headers (8) Test Results

200 OK • 9 ms • 497 B

JSON Preview Visualize

```
1 {
2   "data": [
3     {
4       "_id": "6a4bf53f653d722b2b60a0e2",
5       "name": "Nguyen Van A",
6       "email": "a@gmail.com",
7       "address": "HCM",
8       "phone": "0912345678",
9       "favorite": false,
10      "createdAt": "2026-07-06T18:34:39.746Z",
11      "updatedAt": "2026-07-06T18:34:39.746Z",
12      "__v": 0
13    }
14  ]
15 }
```

- GET <http://localhost:3000/api/contacts/favorite>

http://localhost:3000/api/contacts/favorite

GET http://localhost:3000/api/contacts/favorite

Docs Params Authorization Headers (6) Body Scripts Settings

none form-data x-www-form-urlencoded raw binary GraphQL JSON

1 Ctrl+Alt+P to Ask AI

Body Cookies Headers (8) Test Results

200 OK • 9 ms • 277 B

JSON Preview Visualize

```
1 {
2   "data": []
3 }
```

- GET <http://localhost:3000/api/contacts/<contact-id>>

http://localhost:3000/api/contacts/6a4bf53f653d722b2b60a0e2

GET http://localhost:3000/api/contacts/6a4bf53f653d722b2b60a0e2

Body

1 Ctrl+Alt+P to Ask AI

Body Cookies Headers (8) Test Results

200 OK 7 ms 495 B

```
{
  "data": {
    "_id": "6a4bf53f653d722b2b60a0e2",
    "name": "Nguyen Van A",
    "email": "a@gmail.com",
    "address": "HCM",
    "phone": "0912345678",
    "favorite": false,
    "createdAt": "2026-07-06T18:34:39.746Z",
    "updatedAt": "2026-07-06T18:34:39.746Z",
    "__v": 0
  }
}
```

- PUT <http://localhost:3000/api/contacts/<contact-id>>

http://localhost:3000/api/contacts/6a4bf53f653d722b2b60a0e2

PUT http://localhost:3000/api/contacts/6a4bf53f653d722b2b60a0e2

Body

```
{
  "name": "Nguyen Van A Updated",
  "email": "a@gmail.com",
  "phone": "0987654321",
  "address": "Da Nang",
  "favorite": true
}
```

Body Cookies Headers (8) Test Results

200 OK 8 ms 506 B

```
{
  "data": {
    "_id": "6a4bf53f653d722b2b60a0e2",
    "name": "Nguyen Van A Updated",
    "email": "a@gmail.com",
    "address": "Da Nang",
    "phone": "0987654321",
    "favorite": true,
    "createdAt": "2026-07-06T18:34:39.746Z",
    "updatedAt": "2026-07-06T18:39:56.306Z",
    "__v": 0
  }
}
```

- DELETE <http://localhost:3000/api/contacts/<contact-id>>

http://localhost:3000/api/contacts/6a4bf53f653d722b2b60a0e2

DELETE http://localhost:3000/api/contacts/6a4bf53f653d722b2b60a0e2

Docs Params Authorization Headers (8) Body Scripts Settings

none form-data x-www-form-urlencoded raw binary GraphQL JSON

```
1 {
2   "name": "Nguyen Van A Updated",
3   "email": "a@gmail.com",
4   "phone": "0987654321",
5   "address": "Da Nang",
6   "favorite": true
7 }
```

Body Cookies Headers (5) Test Results

204 No Content · 7 ms · 166 B

Raw Preview Visualize

1

- DELETE <http://localhost:3000/api/contacts>

http://localhost:3000/api/contacts

DELETE http://localhost:3000/api/contacts

Docs Params Authorization Headers (8) Body Scripts Settings

none form-data x-www-form-urlencoded raw binary GraphQL JSON

```
1 {
2   "name": "Nguyen Van A Updated",
3   "email": "a@gmail.com",
4   "phone": "0987654321",
5   "address": "Da Nang",
6   "favorite": true
7 }
```

Body Cookies Headers (5) Test Results

204 No Content · 8 ms · 166 B

Raw Preview Visualize

1

- GET <http://localhost:3000/huhukheckhec/kakaka>

 http://localhost:3000/huhukheckhec/kakaka

Save

Share

DELETE

http://localhost:3000/huhukheckhec/kakaka

Send

DocsParamsAuthorizationHeaders (8)BodyScriptsSettings

noneform-datax-www-form-urlencodedrawbinaryGraphQLJSON

Beautify

```
1 {
2   "name": "Nguyen Van A Updated",
3   "email": "a@gmail.com",
4   "phone": "0987654321",
5   "address": "Da Nang",
6   "favorite": true
7 }
```

BodyCookiesHeaders (8)Test Results

404 Not Found • 5 ms • 316 B

{ }JSON

Preview

Debug with AI

1 {
2 "error": {
3 "message": "Resource not found"
4 }
5 }